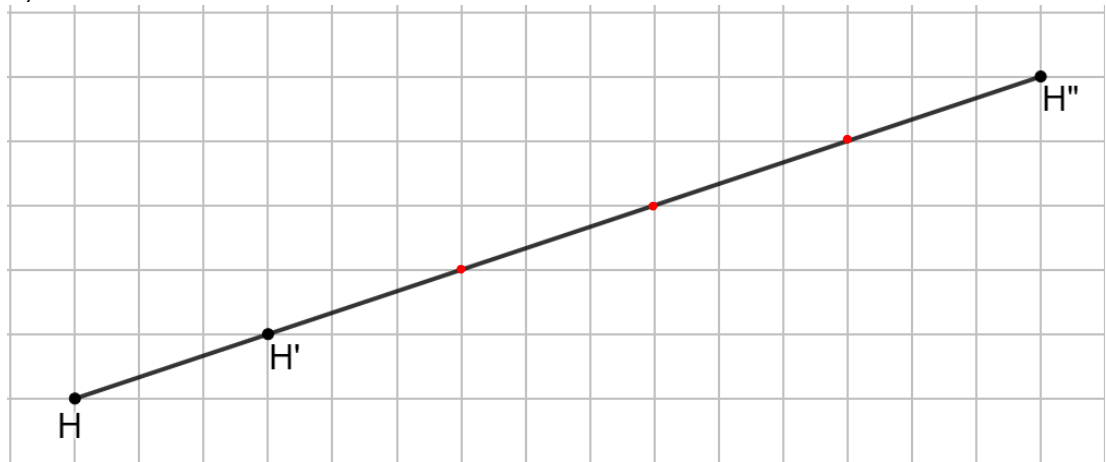


Points H , H' , and H'' are shown.



- 1. Determine the number of copies of $\overline{HH'}$ that will fit on $\overline{HH''}$.
5
- 2. Use the definition of dilation to explain why $\overline{HH''}$ is a dilation of $\overline{HH'}$.
 $\overline{HH''}$ is 5 times the length of $\overline{HH'}$.
- 3. Complete the statement.

Point

☐ H

☒ H'

☐ H''

 partitions

☐ $\overline{HH'}$

☐ $\overline{H'H''}$

☒ $\overline{HH''}$

 so that the ratio of $\frac{HH'}{HH''}$ is

☐ $\frac{1}{4}$

☒ $\frac{1}{5}$

Use the Pythagorean theorem to complete the table.

	Segment	Length
4.	$\overline{HH'}$	$1^2 + 3^2 = c^2$ $10 = c^2$ $c = \sqrt{10}$
5.	$\overline{H'H''}$	$4^2 + 12^2 = c^2$ $160 = c^2$ $c = 4\sqrt{10}$
6.	$\overline{HH''}$	$5^2 + 15^2 = c^2$ $250 = c^2$ $c = 5\sqrt{10}$

7. Complete the statements.

Point H' is

☐

☒

$\frac{1}{4}$
 $\frac{1}{5}$

 of the distance from H to H'' . The scale factor that can be

used to multiply

☐

☐

☒

$\overline{HH'}$
 $\overline{H'H''}$
 $\overline{HH''}$

 by to produce

☒

☐

☐

$\overline{HH'}$
 $\overline{H'H''}$
 $\overline{HH''}$

 is

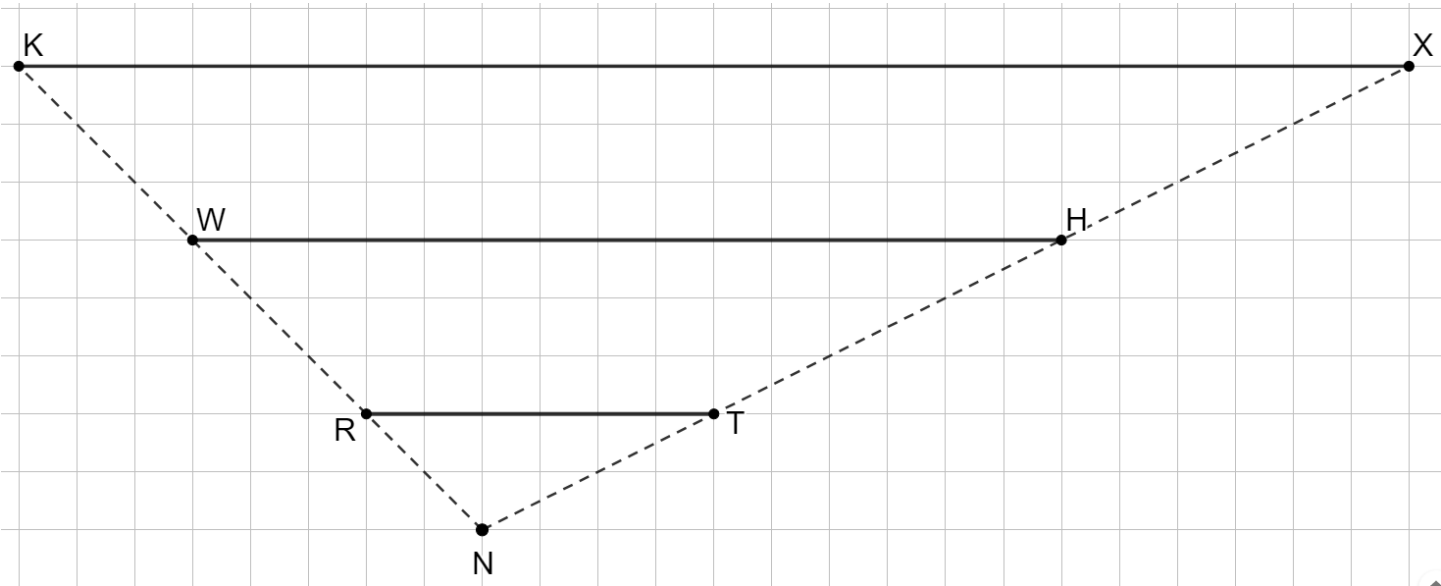
☐

☒

$\frac{1}{4}$
 $\frac{1}{5}$

.

Line segments KX , WH , and RT are shown where $\overline{RT}$ is dilated using point N to create $\overline{WH}$ and $\overline{KX}$.



Determine each of the ratios.

	Ratio	Length
8.	$NR:NW$	$2\sqrt{2}:5\sqrt{2}$
9.	$RT:WH$	$6:15$
10.	$RT:KX$	$6:24$